

EvoLab — Fast-Track Plan

What to build extra, only if you're ahead of schedule

This is a bonus plan, not a replacement. It sits on top of your normal 5-week roadmap. Each week below has one **rule**: don't touch the bonus items until that week's regular, required work is fully done and merged. If the rule isn't satisfied, skip everything below it and just keep working the core plan.

If you can only pick a few bonus items, do them in this order — earlier ones matter more and take less time:

1. Show both the real family tree and the guessed one on screen at the same time
2. Finish the Geographic Isolation experiment
3. Add a second tree-guessing method (Neighbor-Joining)
4. Add mate choice as a selection pressure
5. Let genes on the same chromosome inherit together (linkage)
6. Add the third experiment (sudden environment change)
7. Add a third tree-guessing method (Maximum Parsimony)
8. Let the user trigger events live during the demo

WEEK 1

UNLOCK CONDITION

All 4 people finished their normal Week 1 work, and the team has done the Thursday explain-to-each-other session.

Who	Extra thing to build
Person A	Add a third way genes can combine (codominant) — you'll already have two, this adds the last common one.
Person C	Just start reading about the second tree-guessing method (Neighbor-Joining). Don't build it yet.
Person D	Add a placeholder switch to flip between "real tree" and "guessed tree" on screen, even with fake data for now.

WEEK 2

UNLOCK CONDITION

Your tree-building works correctly on test data, and a small test population runs for several generations without breaking.

Who	Extra thing to build
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Person B	Add a second environmental factor (on top of camouflage) that affects survival.
Person C	Start actually building the second tree method (Neighbor-Joining), alongside the first.
Person D	Make the "real family tree" side of the screen update live, not just the guessed side.

WEEK 3

UNLOCK CONDITION

The full pipeline already works end-to-end — organisms evolve, their real ancestry is recorded, a tree is guessed from their DNA, and you get an accuracy score — and this is confirmed two days early.

Who	Extra thing to build
Person A	Let genes on the same chromosome be inherited together, with an on/off switch, so you can compare results with and without it.
Person B	Add mate choice as a new factor affecting who reproduces.
Person C	Finish the second tree method and show accuracy scores for both methods side by side.
Person D	Get both trees (real and guessed) fully live on screen at once, matching the main pitch.

WEEK 4

UNLOCK CONDITION

Your first experiment has clean results, and your second experiment is at least running (doesn't need to be perfectly tuned yet).

Who	Extra thing to build
Persons A + B	Run the third experiment: a sudden change in the environment mid-run, and observe what happens.
Person C	If you have two tree methods, run both experiments through both methods for stronger results.
Person D	Build the extra chart showing the population adapting over time (fitness landscape).

WEEK 5

UNLOCK CONDITION

All three experiments have final charts/results, and your report and demo are already drafted. This week's bar is stricter — it's your only cushion before submission.

Who	Extra thing to build
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Person A	If gene linkage is done, run one more quick experiment testing how strong the linkage effect is.
Person B	Let the presenter trigger an event live during the demo (like starting an isolation event on command). Skip this if bugs aren't fixed yet — a crash is worse than not having it.
Person C	Only if everything else is solid: try a third tree-guessing method (Maximum Parsimony). This is the most optional item on this whole page.
Person D	If live-triggered events work, show a clear before/after view when one happens — good for the demo recording.

Week 5 has no room for a stalled bonus feature — if in doubt, don't start it.

WEEK 6 (Bonus, only if you somehow finish everything)

UNLOCK CONDITION

All three experiments, the live dual-tree view, and at least one other bonus item above are already done, with a full week still free.

Who	Extra thing to build
Everyone	Run one big combined experiment testing mutation rate, population size, and time together, instead of one at a time.
Everyone	Write a short section explaining specifically when and why the tree-guessing breaks down — this is the real answer to your research question.
Person D	Polish the recorded demo clip — this is genuinely worth the extra time.

The one rule: if the unlock condition for a week isn't met, don't start that week's bonus items — even if one person personally has free time. A half-built bonus feature is worse than not having it, because someone else has to clean it up later. Always finish and merge the required work first.